

Dear Editors,
August 28, 2026

We submit our manuscript "Leakage-Aware Structure-Phenotype Prediction of Mechanism-of-Action Profiles Across a Mapped Drug Cohort" for consideration in the *Journal of Computer-Aided Molecular Design*.

Mechanism-of-action prediction from molecular structure is a central challenge in computational drug discovery, yet benchmarks rarely control for data leakage through identical canonical structures crossing train and test boundaries. We constructed an auditable mapping between 3289 drug-level LISH-MoA records and molecular structures, resolved salts, and assigned collision groups before model evaluation. Under this leakage-controlled protocol, all molecular representations (GIN, GIN-TFP, GIN-TNE, ChemBERTa) produced near-chance AUROCs (0.501–0.508), well below the phenotype-only reference (0.636).

The key contributions are: (1) a collision-group-aware evaluation framework that prevents identical structures from contaminating test folds; (2) a systematic comparison showing that molecular structure alone does not improve MoA-associated label prediction on this drug-level cohort; and (3) an honest negative result demonstrating that calibrated log loss can accompany non-discriminative ranking.

The manuscript is original, has not been published elsewhere, and is not under consideration by another journal. All authors have approved the manuscript and agree with its submission.

We suggest the following potential reviewers with relevant expertise in molecular representation learning and benchmark design. No conflicts of interest exist with any suggested reviewer.

Thank you for your consideration.

Sincerely,

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